AIML Minor PROJECT

Product Review Sentiment Analysis using TF-IDF and Machine Learning

Predicting whether a product review is Positive or Negative

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Manual review reading is too slow at scale

Stores receive thousands of product reviews. Manual reading takes significant time.

This project builds a system that reads a review and automatically labels its sentiment as Positive or Negative.

Thousands

of reviews arrive at stores

Too slow

to read manually at scale

PROJECT GOAL

Predict sentiment

Classify each review as **Positive** or **Negative**.

Pipeline: clean, split, vectorize, classify, evaluate.

Classical machine learning only; no deep learning.

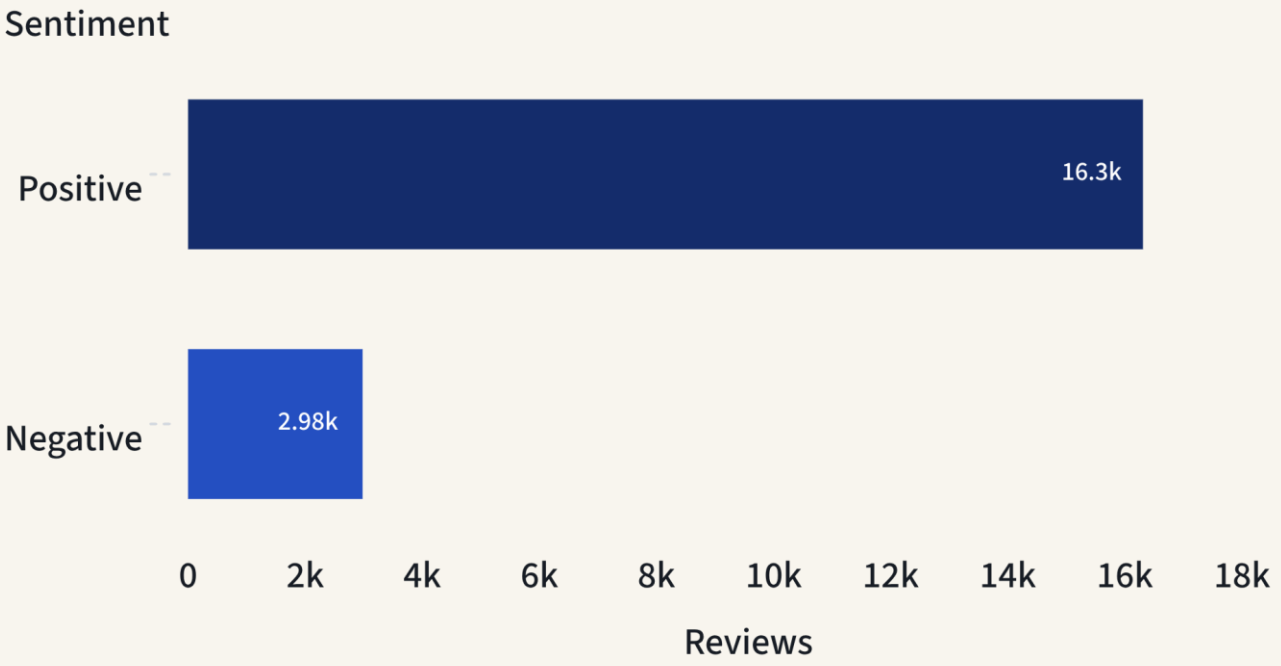
predicts the sentiment of a new, user-entered review through a simple web interface.

SO WHAT

A repeatable classifier turns a high-volume reading task into a measurable prediction problem.

The dataset is strongly imbalanced toward Positive reviews

REVIEW SENTIMENT DISTRIBUTION



Positive 84.55% · Negative 15.45%

SOURCE

Amazon Fine Food Reviews from Kaggle.

LABELS

4–5 stars Positive; 1–2 stars Negative. 3-star reviews removed.

CLEAN SAMPLE

20,000 sampled; **711** duplicates removed; **19,289** remain.

SPLIT

80% train, 20% test, stratified.

SO WHAT

The model must be judged beyond overall accuracy because Positive reviews dominate the sample.

The pipeline turns raw text into a sentiment prediction

1. Raw review — original customer text enters the pipeline.
2. Text cleaning — remove HTML, URLs, punctuation, and digits; lowercase.
3. Train/test split — 80% train and 20% test.
4. TF-IDF fitted on training data only.
5. Logistic Regression.
6. Sentiment plus confidence.
7. Streamlit app — expose the result through a working web app.

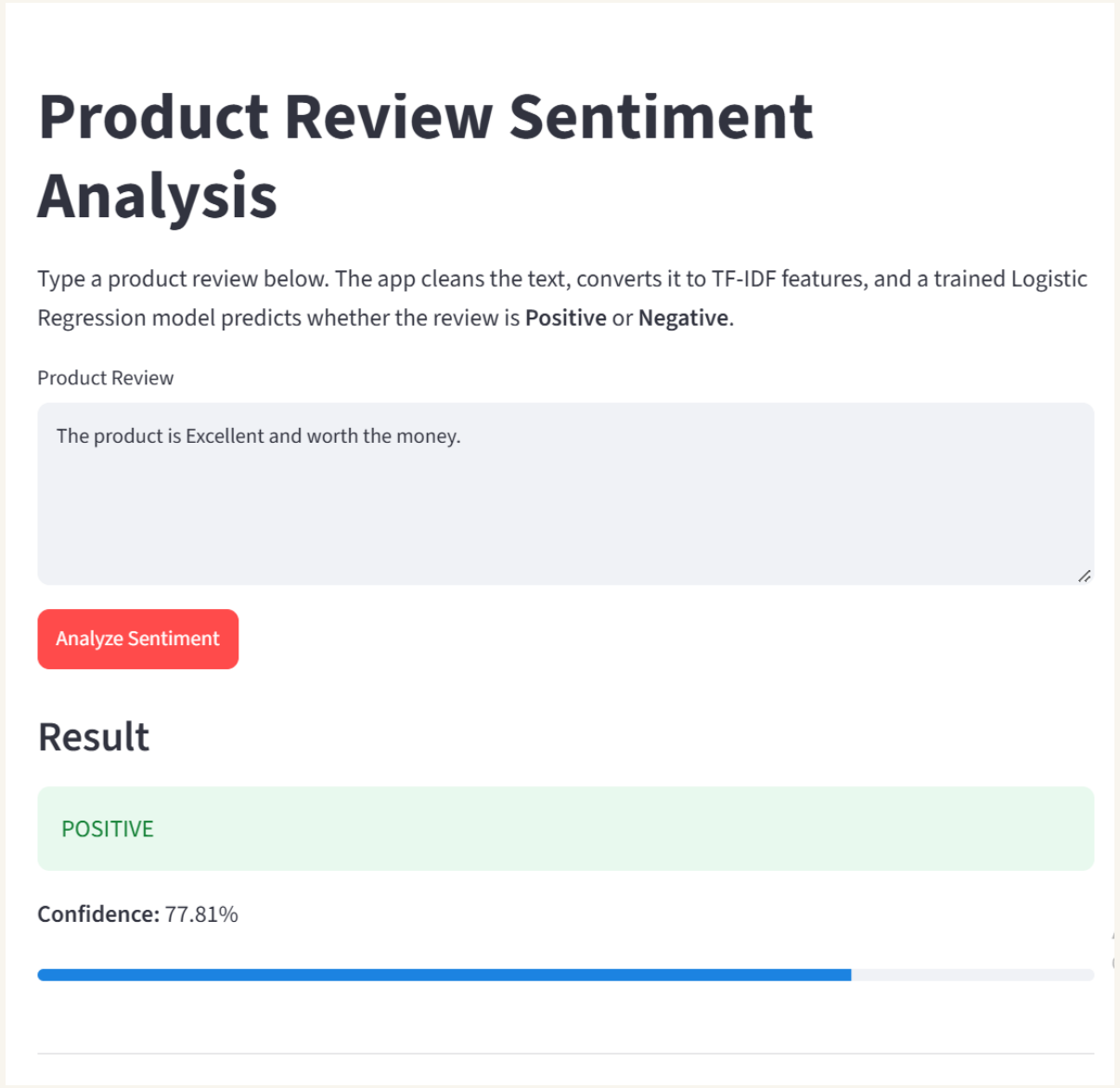
METHOD DETAILS

- Keep negation words: “not,” “no,” and “never.”
- Use 5,000 TF-IDF features with unigrams and bigrams.

The workflow preserves the text signal, converts it into features, and ends with an interpretable prediction.

SO WHAT

The pipeline turns raw text into a sentiment prediction.



Each design choice protects signal and reduces leakage

Split before TF-IDF

Fit the vectorizer on training data only to avoid data leakage.

Keep negation

Custom stopwords keep “not” and “no.”

Use bigrams

Bigrams preserve phrases like “not good.”

Choose Logistic Regression

It is simple, fast, and explainable.

Balance the classes

`class_weight="balanced"` supports the smaller Negative class.

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The model performs well overall, but Negative precision is weaker

TEST-SET RESULTS · 3,858 REVIEWS

Metric	Positive	Negative	Overall
Precision	**0.9695**	0.6477	—
Recall	0.9163	0.8423	—
F1	0.9422	0.7323	—
Accuracy	—	—	**0.9049**
Macro F1	—	—	**0.8372**
Dummy always Positive	—	—	0.8455

CONFUSION MATRIX · ACTUAL / PREDICTED

MATRIX	Negative	Positive
Negative	**502**	94
Positive	273	**2989**

Saved confusion matrix image placeholder
Replace this marked area with the saved confusion matrix image.

SO WHAT Accuracy is strong; the Negative class needs attention: precision is 0.6477 despite recall of 0.8423.

The live demo shows confidence, while Logistic Regression beats Naive Bayes on this split

STREAMLIT DEMO PLACEHOLDER

Text box → Analyze Sentiment → result and confidence

“Excellent product. The quality is amazing.”
Positive, 97.62%.

“The product is not good.”
Negative, 95.98%.

“Good quality but delivery was very late.”
Positive, 84.33%.

Placeholder: Streamlit app screenshot

MODEL COMPARISON

Metric	Logistic Regression	Multinomial Naive Bayes
Accuracy	**0.9049**	0.8730
Negative recall	**0.8423**	0.1963
Macro F1	**0.8372**	0.6266

Naive Bayes labeled most reviews Positive. This is one split, not a general ranking.

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The pipeline works today, with clear next steps

Tested pipeline from raw text to working web app.

What it does today

Scope covers food reviews with star-rating labels. All **15 end-to-end tests passed.**

What remains limited

Negative precision is weak, only two classes are used, and mixed reviews are forced into one class.

FUTURE

Next steps

- Use the full dataset.
- Apply cross-validation.
- Add a Neutral class.
- Support more categories.

Thank you

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